

COURTSIDE: Pricing the Value of Speed in In-Play Tennis Prediction Markets

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Abstract. We study who profits from speed in Polymarket’s in-play tennis moneylines (13,084 matches, public tapes). A walk-forward fast tier of wallets trading within 3 s of a score move earns after fees in every in-sample and held-out month; other takers lose, and copying them 3 s later loses. At the fast tier’s own fills a frozen book (v2) has a Sharpe ratio of 14.5 in sample and 6.7 on a burned hold-out, but turns negative when fees double. We then price the speed a computer-vision (CV) trader needs (simulated; assumed feed latency (licensed feed not purchased); parameters measured). Pre-registered, it breaks even at a 1.0–1.1 s feed delay and earns +\$15, +\$4 and −\$13 a day held out at 0.5, 1 and 3 s; a post hoc stamp-lag inference gives +\$57 at 1 s. A replay on real books loses. Real money: none.

JEL: G14, G12, C58. **Keywords:** prediction markets; latency arbitrage; in-play betting; tennis; computer vision; market microstructure.

1 Summary

- **Strategy and edge.** Call the point with CV and lift the stale resting quote inside the venue’s 1 s taker delay. The book reprices 1.2 s *before* the official stamp; the fast tier earns +0.93 / +0.75¢ a share after fees (IS / OOS) in 9/9 and 3/3 months; other takers lose.
- **Headline, burned OOS** (Table 1). v2 at fast-tier fills: Sharpe 6.7 [1.9, 12.3], +0.60¢ [0.09, 1.13]. CV at a 1 s feed, pre-registered: +\$4/day, −0.38¢ [−2.08, 1.21]: break-even (the call must reach the venue 0.89 s before the umpire’s stamp). Post hoc (stamp lag 3.14 s; trades selected on outcomes, not ex ante): +\$57/day. Real books: −0.94¢. Our pipeline, frame to order on own footage: 54 ms.
- **What failed.** Fees ×2 OOS (−0.34¢); the blind unseen-market, maker and v3 tests; table tennis; v2 OOS net of a central feed licence (−\$75/day). Real money: none.

2 Economic hypothesis

Win probability is a known function of the score. The leverage of point k ,

$$\Delta(k) = \Pr(\text{win} \mid \text{point } k \text{ won}) - \Pr(\text{win} \mid \text{point } k \text{ lost}), \quad (1)$$

follows from the point-level Markov chain (Klaassen and Magnus, 2001); mean $|\Delta|$ is 5.7%. Each point moves fair value by a known amount at a known instant, and the news reaches traders in tiers (Fig. A1a); stale makers are picked off by whoever learns first (Glosten and Milgrom, 1985; Budish et al., 2015; Aquilina et al., 2022). The rent is being competed away: the fast tier grew from 4 to 131 wallets as its edge fell 0.20¢ a month ($t = -3.98$). In-play mispricing is known (Croxson and Reade, 2014; Brown, 2012); new here: a wallet-level map of the latency tier and a price on each second.

Reproduce: bash reproduce.sh; paper: python scripts/build_paper.py (<https://github.com/OjasMishra32/computervisionpredictivetabletennismoneymachineouuushiii>). Paper only: no real money, no licensed feed, no match footage and no court camera.

Table 1. Headline metrics. (A) v2 at the fast tier's own fills. (B) The CV strategy at three assumed feed latencies: pre-registered, it breaks even near 1 s; the sign turns on one unmeasured number, the stamp lag L .

A. Fast-tier fills	v2 (frozen)		v2-safe (net cap 50)	
	IS	OOS	IS	OOS
Sharpe	14.5	6.7	15.7	9.3
95% CI	[11.9, 17.4]	[1.9, 12.3]	[13.1, 18.6]	[4.5, 15.6]
Deflated Sharpe	0.997	0.075	1.000	0.290
Net ¢ per share	+1.38	+0.60	+1.33	+0.69
Return / vol, a year	253% / 17.5%	148% / 22.2%	239% / 15.2%	170% / 18.4%
Max DD / worst month	-2.0% / +\$2,543	-2.1% / -\$434	-1.6% / +\$1,499	-1.9% / -\$187
Turnover (×/yr) / skew	93 / +0.53	129 / +0.33	90 / +0.61	129 / +0.17
B. CV strategy (simulated)	Pre-registered $L = 2.0$ s		Post hoc $L = 3.14$ s	
	IS	OOS	IS	OOS
Feed latency V (assumed)				
0.5 s best case: \$/day	+\$28	+\$15	+\$133	+\$81
Sharpe / net ¢ per share	4.0 / +0.61	2.1 / -0.02	15.5 / +1.21	12.2 / +0.82
¢, 95% CI	[0.08, 1.14]	[-1.22, 1.09]	[1.00, 1.42]	[0.39, 1.25]
1 s base case: \$/day	+\$15	+\$4	+\$94	+\$57
Sharpe / net ¢ per share	2.1 / +0.40	0.3 / -0.38	11.9 / +1.11	8.8 / +0.66
¢, 95% CI	[-0.32, 1.08]	[-2.08, 1.21]	[0.85, 1.38]	[0.10, 1.19]
3 s requirement: \$/day	-\$13	-\$13	-\$6	-\$11
Sharpe / net ¢ per share	-2.6 / -1.35	-2.6 / -1.69	-1.4 / -0.79	-2.4 / -1.47
¢, 95% CI	[-3.15, 0.37]	[-4.85, 1.32]	[-2.26, 0.60]	[-4.45, 1.34]
Break-even feed delay V , s	1.09	1.01	2.23	2.14
At 1 s: clocks read per point, \$/day	-\$17	-\$17	-\$17	-\$17
no early CV calls, \$/day	+\$8	-\$13	+\$88	+\$49
live engine's calls, \$/day	+\$17	+\$4	+\$98	+\$57

Notes: Sharpe: daily P&L, zero-filled calendar days, $\times\sqrt{365}$; CI: stationary bootstrap; deflated Sharpe at 3,386 trials (3,410 for v2-safe), lowest of three variance sources. OOS: burned, non-blind; v2-safe was fixed after v2's OOS losses and failed blind. B: assumed feed latency (licensed feed not purchased); parameters measured; 20 seeds a cell; trades are points that later repriced $\geq 4\text{¢}$, selected on outcomes (not ex ante); timing per tournament, adopted after the first P&L ($V3$); the post hoc lag assumes courtside humans, 95% CI 2.23–3.22 s; per point, no correct call fills at 1 s; CV recall is the offline rule's (look-ahead feature). PREREG.md was written 17:25 UTC, before the first P&L.

H6. Universe: Polymarket ATP/WT/Challenger moneylines. Behaviour: wallets trading 0–3 s after a score move. Horizon: 30 s markout. Counterparty: the resting maker who has not yet cancelled a stale quote. Persistence: physics and data licences decide who learns first; the 1 s taker delay lets makers cancel (the cost D in Eq. 2). Prediction: fast wallets chosen on past months keep a positive net markout each month and out of sample. Falsifier: ≤ 0 in over a third of months, or out of sample. Trail: H1–H4 committed before any result (7232986); H6 frozen before the OOS opened.

3 Data and universe

13,084 resolved ATP/WT/Challenger singles moneylines with $\geq \$5,000$ volume, Oct 8, 2025 to Oct 3, 2026, public books and tapes (Polymarket, 2026a). IS: the first 80% of matches by start; OOS: the last 2,617 from 2026-08-25 14:15 UTC (burned for v2 and the CV simulation). A blind test on

Table 2. *The best version of each component, its best honest result, and how it was chosen.*

Component	Best version: best honest result	Selection
v2 sizing	G_50pct_net100: Sharpe 14.5 / 6.7	walk-forward, 55 policies
CV, table tennis	live causal engine: 4 / 41 misses called early (median 162 ms), 0 wrong	tested once; spin model not adopted
CV, tennis	spin-aware tracker: landing error 0.57 cm vs 5.76 cm at 200 ms	simulated physics; no tennis video
Engine, pipeline	L4 GPU: 119.9 fps, call-ready in 4.6 ms p50; frame to order 54 ms	fastest backend that keeps up
CV rule	frozen v3: -0.52¢ blind	360 IS variants

11,307 smaller markets checks survivorship; 6 tapes may be truncated, 2,889 early matches pay no fee, 2.89% settle 50/50 (kept), and tape stamps lag 1.98 s. Timing: ESPN, WTA (ESPN, 2026; WTA, 2026); vision: OpenTTGames (Voeikov et al., 2020), BlurBall and WASB weights (Gossard et al., 2026; Tarashima et al., 2023).

4 Methodology

Fast tier and v2. A detector fires on a $\geq 4\text{¢}$ move of the 10 s VWAP against the prior 60 s. Each month a wallet qualifies on past months only (≥ 30 prints 0–3 s after a detection, ≥ 10 matches, $t > 3$). The frozen book v2 copies their prints, sized by resolution risk, at most \$1,000 an order and 100 shares net per match, held to resolution, at the fast tier’s own prices: *the opportunity at their speed, not our execution* (components: Table 2).

CV and the latency budget. Frame to unsigned order takes 54 ms; with a simulated 1 s feed, 65 ms network and the 1 s venue hold an order is executable at 2,119 ms, inside the organiser’s 3 s bar (Fig. 3; paper, no order sent). It wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (2)$$

where V is the feed delay, δ our pipeline and network, D the 1 s taker delay, ℓ the call lead, L the stamp lag and R the reprice relative to the stamp (measured, $n = 482$). The trade set is the historical $\geq 4\text{¢}$ detector set, *selected on outcomes, not ex ante*; the replay calls every point. L is unmeasured; Table 1 shows every reading. Only $L + R$ enters, so $V = 1$ s at the post hoc $L = 3.14$ s equals a camera 0.14 s ahead of the point at $L = 2.0$ s.

Costs. Each match’s own taker fee, $f(q) = r q (1 - q)$ per share = 10,000 $r (1 - q)$ bps of notional, $r = 5\%$ today; spreads are paid as traded; Table 3 doubles them. **Validation.** Walk-forward inside IS; the fee in force; every read of held-out data logged.

5 Results

The fast tier beats other takers every month and copying it 3 s later loses (Fig. 1a). v2 earns in both periods (Table 1, Fig. 1b); out of sample the edge is gone at $\frac{1}{2}$ tick worse entry (+0.10¢) and negative with fees doubled (−0.34¢). **Sharpe above 3.** We assumed a bug and found one: fixing the onset look-ahead (D9) cut IS Sharpe from 16.8 to 14.5. The rest is about 270 small, nearly independent bets a day. The deflated Sharpe (Bailey and López de Prado, 2014) is 0.997 IS but 0.075 OOS: 40 days cannot rule out luck. **Latency scenarios.** Pre-registered, the CV trader makes +\$15, +\$4 and −\$13 a day held out at 0.5, 1 and 3 s (Table 1B, Fig. 2); post hoc, +\$81, +\$57 and −\$11, but at its interval’s low end only +\$8. The replay calls every point ex ante and loses in 36 of 36 settings (Fig. A6b): the gap is the value of knowing in advance which points matter, which no tested rule supplies yet.

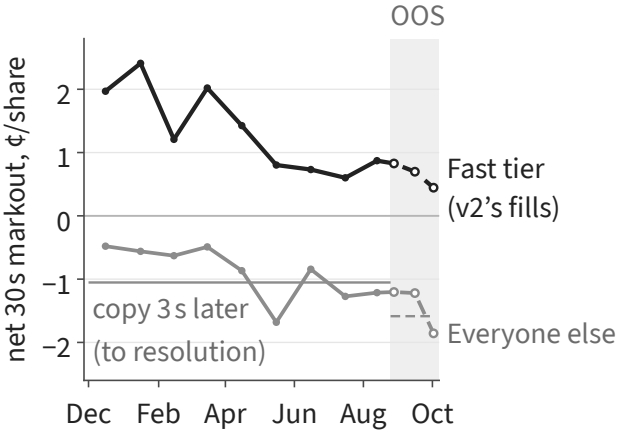
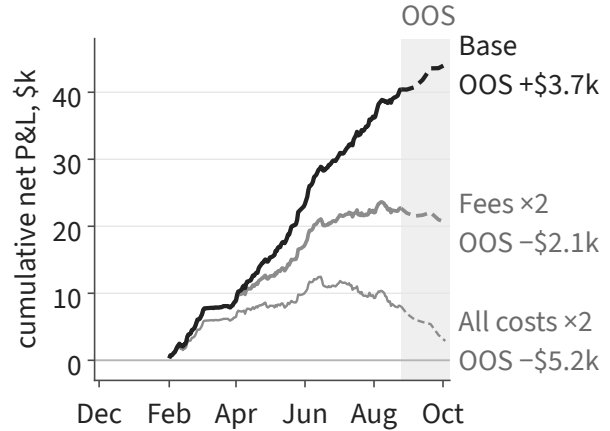
a Only the fast tier earns; copying late loses**b** v2 earns; doubled costs erase the OOS gain

Figure 1. *The edge exists at the fast tier's fills.* (a) Monthly net 30 s markout (copying 3 s later: held to resolution). (b) v2 net P&L. Filled / solid = IS, hollow / dashed = burned OOS.

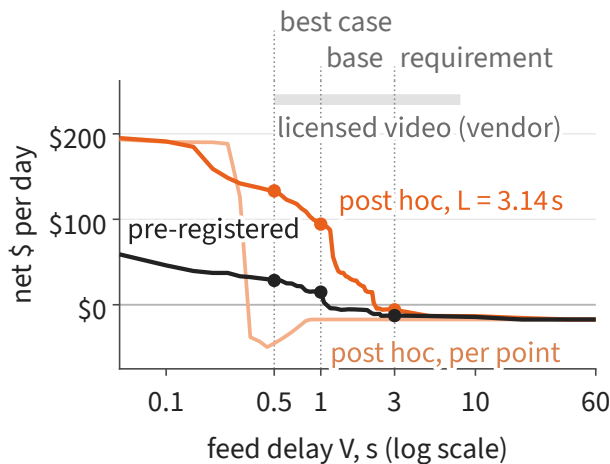
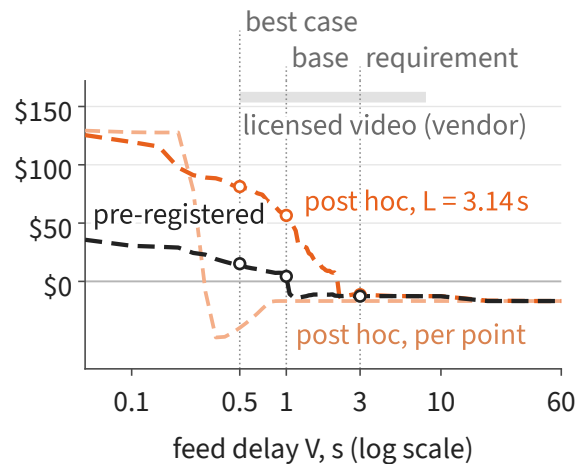
a IS: zero profit at 1.1–2.2 s of delay**b** OOS: zero profit at 1.0–2.1 s of delay

Figure 2. *Each second of feed delay costs the edge.* CV strategy, (a) IS and (b) burned OOS, by stamp-lag reading; dots: the scenarios of Table 1B; grey bar: licensed video (vendor-stated). Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

6 Risk management

- **Limits, in code; de-risking, set in advance.** \$1,000 per order, 100 shares net per match, zone 0.05–0.95: one match can lose at most \$95 plus fees. Kill switches: stale feed >2s, vision >1s, latency above its rolling p95; a \$1,000 daily stop. Halve size if the trailing 30-day edge is below 0.3¢; stop at ≤ 0 , a 5% drawdown, or a fee or delay change.
- **CV model risk.** Wrong calls always fill; MISS calls trade only at confidence ≥ 0.95 , and a confirmed false call halts the day. Phantom calls outside play (about 30 an hour) would erase the 1s OOS P&L at 2.5 (pre-registered) or 32 (post hoc) a day; a rally-state gate is coded but not yet evaluated.
- **Tail, venue, legal.** The UMA oracle settles (Polymarket, 2026c); US persons are close-only (Polymarket, 2026b); courtsiding breaches ticket terms (International Tennis Federation, 2026). Factors (Fama and French, 1993; Carhart, 1997): $\alpha t = 8.48$, $\max |t| = 1.36$.

Table 3. *Robustness: every blind or ex-ante test of a tradable book failed.*

Test, net ¢ per share	IS	OOS or blind	Verdict
v2, ½ / 1 tick worse entry	+0.88 / +0.38	+0.10 / -0.40	≈0 at ½ tick
v2, fees ×2 / all costs ×2	+0.77 / +0.27	-0.34 / -0.84	fail OOS
v2, 11,307 unseen markets (blind)	+2.02	+1.22 [-0.19, 2.65]	fail
v2-safe, unseen markets (blind)	+2.11	+1.21 [-0.02, 2.43]	fail
v1 copy fast tier (OOS opened once)	+1.16	+0.57 (-\$36,056)	fail
Maker v1 (blind), per fill	+2.02	+1.87 [-0.14, 3.86]	fail
CV v3 rule, unseen markets (blind)	-0.22	-0.52	fail
CV replay on 9 real books, V = 1 s		-0.94 [-1.28, -0.58]	loses
ex-ante swing ≥4¢ (exploratory)		-0.95 [-1.64, -0.40]	loses
v2 after fixed costs, \$/day	+\$29	-\$75	fail OOS
PBO / bootstrap $P(SR \leq 0)$	15%	0.0023	pass
Forward test and live paper session: not run (submitted before the forward window closed)			

Notes: **4,219 variants tried**; held-out data read 77 times; rules changed after a look: 2 for v2, 13 for the CV simulation. The headline stamp-lag reading and $V = 1$ s were chosen after looking (Table A6).

Frame to executable order in 2.12 s; our part 54 ms

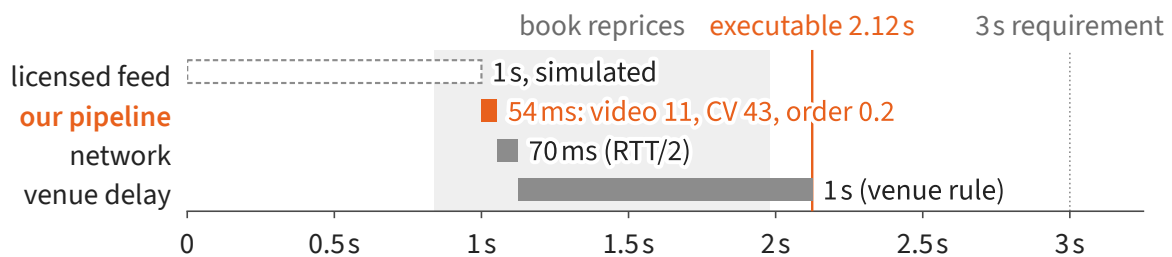


Figure 3. *We can trade inside 3 s.* Stage means of 24 traces: own footage over WebRTC into the CV engine on a laptop (on an L4 GPU: 4.6 ms; WebRTC video leg alone 4–8 ms), a simulated 1 s feed, the measured network and the 1 s venue hold; order built, not sent. Shaded: when the book reprices.

7 Liquidity and capacity

v2 trades \$7,182 a day, 0.07% of match volume; its OOS CI excludes zero only up to 1× size (\$22,754) and 5× loses. The CV book at the pre-registered lag has Sharpe 2.7 / 0.8 (IS / OOS) at its smallest size: no capacity. Post hoc, its Sharpe halves at \$40,000 of capital OOS and \$73,000 IS (Fig. A4); \$/day peaks at \$177 on \$83,000 OOS (Sharpe 1.3). It trades 0.22% of in-play volume but 9% of the fast tier's 0–3 s volume: the seconds after each point bind. The cheapest data stack (\$42 a day) needs \$14,000–\$16,000 of capital post hoc; at 1 s the book can pay at most \$1,644 a month for data post hoc and \$55 pre-registered, against quotes of \$1,250–\$10,000. At today's fee COURTSIDE prices speed; it is not yet a business.

8 Limitations and next steps

One number, the stamp lag L , moves the break-even one for one (Eq. 2); its post hoc value assumes human reaction, but the 1 s hold dominates ($L \geq 2.85$ s even for a 20 ms bot). One session with a licensed feed and test orders measures L and whether delayed orders are released on a 1 s clock. Licensing that feed is a purchase decision, not yet a deployment: sub-second video is sold to licensed sportsbooks, and at 1 s the book covers the cheapest data quote only if $L \geq 3.0$ s. CV accuracy is from table tennis; tennis is simulated; no order was ever sent.

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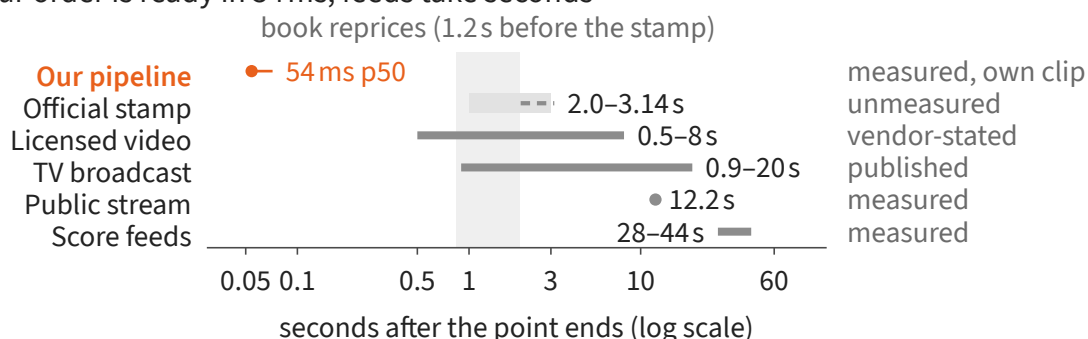
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Disclosure. No real money was used. Simulated components are labelled. The authors hold no licensed data feed, no match footage and no court access. Authors: Ojasva Mishra, Yoan Exposito, Rafael Penhas, Ian Hoang (University of Florida).

Appendix

Everything needed for scoring is in the five main pages; this appendix gives the remaining figures, the full grids, every formula with a worked example, everything we tested, and the logs. Every appendix float repeats its label.

a Our order is ready in 54 ms; feeds take seconds



b Live engine: 4 of 41 misses called early, none wrong

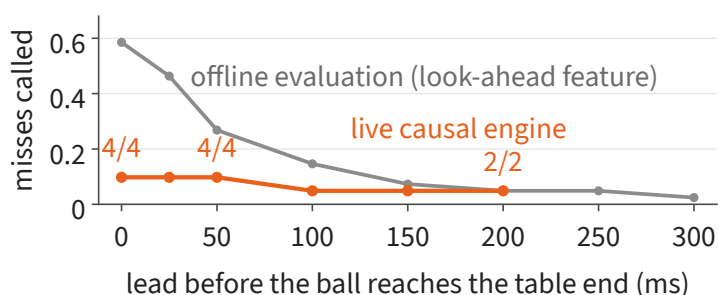


Figure A1. *Who sees the point first.* (a) Delay after the point ends (log): our pipeline (frame to order-ready, own clip over WebRTC on a laptop; p50 dot, p99 whisker); official stamp (dashed: the pre-registered 2.0 s to the post hoc 3.14 s; grey: the cited 1–3 s band); vendor-stated and published ranges; measured public stream and score feeds (ESPN 28.2 s, Polymarket sports 30.0 s, WTA 44.1 s behind the book). Shaded: when the book reprises. (b) Early out-calls on 41 held-out 120 fps table-tennis misses: the live causal engine (orange; labels: right / called) and the offline evaluation, which used a look-ahead feature (a whole-flight median) and kept deciding after a far-side bounce (grey; precision 1.0 up to 100 ms). The causal engine is the deployable number (engine/README.md). Source: results/{e2e,webrtc,tracking,engine}, research/v2/latency.

Table A1. *Everything we tested: each lens, what it was, its best result and the verdict.* IS = in sample; OOS = held out; ¢ = net cents per share.

Lens	What it was	Best result	Verdict
H1 follow the jump	taker in the direction of a $\geq 4\text{¢}$ move, 30 s hold	OOS -2.08¢ $[-2.17, -1.98]$; 20 of 20 IS variants below zero	rejected
H2 favourites	buy favourites priced 0.85–0.97 in play	OOS $+0.80\text{¢}$ $[-1.36, 2.66]$	no edge
H3 early CV call	table-tennis MISS call 50 ms before contact, precision ≥ 0.95	live causal engine 4 of 41 misses, none wrong; offline evaluation (look-ahead feature) 11/11	pass on precision; recall low
H4 score feeds	public score feeds lag the book	median lead 44.5 s ($n = 75$)	confirmed

Lens	What it was	Best result	Verdict
H5 quote after jumps	maker quotes after a move	OOS +0.16¢ [-0.49, 0.87]	n.s.
H6 fast tier	walk-forward wallets trading 0–3 s after a move	+0.93 / +0.75¢ net 30 s; 9/9 and 3/3 months	confirmed
v1	copy the fast tier, \$ sizing	OOS +0.57¢, −\$36,056	failed (blind)
Exit lens	maker exit instead of holding (902 variants)	IS, 1 s/5% regime +0.53¢ [0.42, 0.64]; live fills adverse (27% filled in 30 s, mid then +5.6¢)	not adopted
Sizing lens	55 sizing policies, walk-forward	G_50pct_net100: Sharpe 14.5 IS	adopted (v2)
Selection lens	which fast-tier trades to take (341 variants)	nested walk-forward +1.76¢ [1.36, 2.16] vs +1.16¢ (IS)	IS only
Cross-market	side markets vs the moneyline (68 variants)	leaning maker +2.73¢ [1.73, 3.70] (IS); taking stale side quotes: no edge	IS only; tiny capacity
Kalshi lead-lag	Kalshi vs Polymarket on the same points	Kalshi first in 68.9% of 106,347 repricings (median 5 s)	lag mostly mechanical
Latency and block lag	when the book, stamp and tape move	book 1.2 s before the stamp ($n = 482$); tape stamps 1.98 s late	measured
v2	frozen fast-tier book at its fills	Sharpe 14.5 / 6.7; OOS +0.60¢ [0.09, 1.13]	headline; fees ×2 fail
v2 unseen markets	11,307 never-examined markets (blind)	OOS +1.22¢ [-0.19, 2.65]	fail
v2-safe	net cap 50, fixed after v2's OOS losses	Sharpe 9.3 burned OOS; blind +1.21¢ [-0.02, 2.43]	fail (blind)
Maker v1	pre-registered maker book (blind)	+1.87¢ [-0.14, 3.86] per fill, −\$379	fail
Tier-0 camera	CV trader with an in-venue camera ($V = 0$)	+\$89 / +\$46 a day (pre-registered)	simulated, no camera held
Latency sweep	V 0–60 s, six stamp-lag readings (944 cells)	pre-registered break-even 1.01–1.09 s	Table 1
Tier-0 v3	rule optimised on 360 IS variants	blind −0.52¢	fail
Replay, all points	9 matches' real books, every point called	−0.94¢ [-1.28, −0.58] at $V = 1$ s; 36 of 36 settings below zero	loses
Replay, ex-ante swing	only points with a large Markov swing (exploratory)	−0.95¢ [-1.64, −0.40]; 9 of 9 cells below zero	loses
Table tennis markets	TT1–TT5 on Polymarket table tennis	no fast wallet qualifies; spread 94¢	untestable
Spin model, table tennis	spin-aware early call, held-out footage	10/12 right at 50 ms vs the frozen model's 11/11 (both offline)	not adopted
Tennis tracker	spin-aware landing model (simulated physics)	0.57 vs 5.76 cm at 200 ms; OUT precision 1, recall 0.98	simulation only
GPU engine	streamed causal engine on an L4	119.9 fps; 4.6 ms p50; 0 frames dropped	measured

Lens	What it was	Best result	Verdict
WebRTC pipeline	our clip over WebRTC into the engine	46 ms p50, 67 ms p99 frame to call; video leg 4–8 ms	measured, own clip
End to end	frame to unsigned order on a live book	54 ms ours; 2,119 ms with a 1 s feed vs 3,000 ms	pass (paper)
Capacity	size grid, CV and v2	post hoc Sharpe halves at \$40,000–\$73,000; pre-registered: none	small
Factor regression	FF three factors and momentum	$\alpha t = 8.48$; $\max t = 1.36$	factor-neutral
Rigor pack	DSR, PBO, block bootstrap	DSR 0.997 / 0.075; PBO 15%; $P(SR \leq 0) = 0.0023$	luck not excluded OOS

Calculations

Every statistic in the paper, its formula and one worked example with our numbers (source keys in results/paper/numbers.json).

C1. Annualised Sharpe. $SR = \sqrt{365} \bar{d} / s_d$, where d_t is the daily net P&L on every calendar day of the period (zero on days without trades) and s_d its standard deviation ($n - 1$). v2 IS: $\bar{d} = \$196.24$, $s_d = \$258.87$, $T = 206$ days, so $SR_d = 0.758$ and $SR = 0.758 \times 19.105 = 14.48$. OOS: $\$92.19 / \$264.15 \times 19.105 = 6.67$.

C2. Sortino. $\sqrt{365} \bar{d} / DD$, $DD = \sqrt{\frac{1}{T} \sum_t \min(d_t, 0)^2}$. v2 OOS: $DD = \$125.64$, so $\sqrt{365} \times \$92.19 / \$125.64 = 14.0$.

C3. Maximum drawdown. $MDD = \min_t (E_t - \max_{s \leq t} E_s)$, $E_t = \sum_{u \leq t} d_u$, reported as a share of capital. v2 OOS: $-\$469 / \$22,754 = -2.06\%$.

C4. Probabilistic Sharpe. $PSR(SR^*) = \Phi\left((\hat{SR} - SR^*)\sqrt{T-1} / \sqrt{1 - \gamma_3 \hat{SR} + \frac{\gamma_4 - 1}{4} \hat{SR}^2}\right)$ on daily SR, skew γ_3 and kurtosis γ_4 (Bailey and López de Prado, 2014). v2 OOS: $\hat{SR} = 0.349$, $T = 40$, $\gamma_3 = 0.326$, $\gamma_4 = 3.26$: $PSR(0) = 0.987$.

C5. Deflated Sharpe. $DSR = PSR(SR_0)$ with $SR_0 = \sqrt{V}\left((1 - \gamma)\Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma\Phi^{-1}\left(1 - \frac{1}{Ne}\right)\right)$, γ the Euler–Mascheroni constant and N the trial count. v2 OOS, $N = 3,386$, null variance $V = 1/(T - 1)$: Φ^{-1} terms 3.436 and 3.698, $SR_0 = 0.574$ daily (10.97 annualised), $DSR = 0.075$. Table 1 reports the lowest DSR over three variance sources.

C6. Probability of backtest overfitting. CSCV (Bailey et al., 2017): split the daily P&L of 24 variants into $S = 16$ blocks; for each of the $\binom{16}{8} = 12,870$ halvings pick the IS-best variant and take the logit $\lambda = \ln \frac{\omega}{1-\omega}$ of its OOS rank ω ; PBO = $\Pr(\lambda \leq 0)$. v2-safe grid: PBO = 15% (median $\lambda = 1.27$).

C7. Stationary block bootstrap. (Politis and Romano, 1994): resample the daily series in blocks of geometric length (mean 5 days), 10,000 draws; percentile CI. v2 OOS Sharpe: [1.9, 12.3], standard error 2.65, $P(SR \leq 0) = 0.0023$.

C8. Factor regression. $r_t = \alpha + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_U \text{Mom}_t + \varepsilon_t$ on daily return on capital, Newey–West (5 lags) (Fama and French, 1993; Carhart, 1997; Newey and West, 1987). v2 IS (138 days): $\alpha = 0.801\%$ a day ($t = 8.48$); $\beta_M = 0.173$ ($t = 1.36$), $\beta_S = 0.085$, $\beta_H = 0.044$, $\beta_U = 0.028$; $R^2 = 3.3\%$.

C9. Taker fee. $f(q) = r q(1 - q)$ per share at price q ; per dollar of notional $f/q = r(1 - q)$, i.e. 10,000 $r(1 - q)$ bps. At $r = 5\%$: $q = 0.5$ pays 1.25¢ a share (250 bps); $q = 0.8$ pays 0.80¢ (100 bps).

C10. Markov point leverage. Eq. 1, with the match won probability computed exactly by recursion over points, games, tiebreaks and sets (src/markov.py). Two even players who win 0.645 of

Held-out miss called 408 ms early offline, 325 ms live

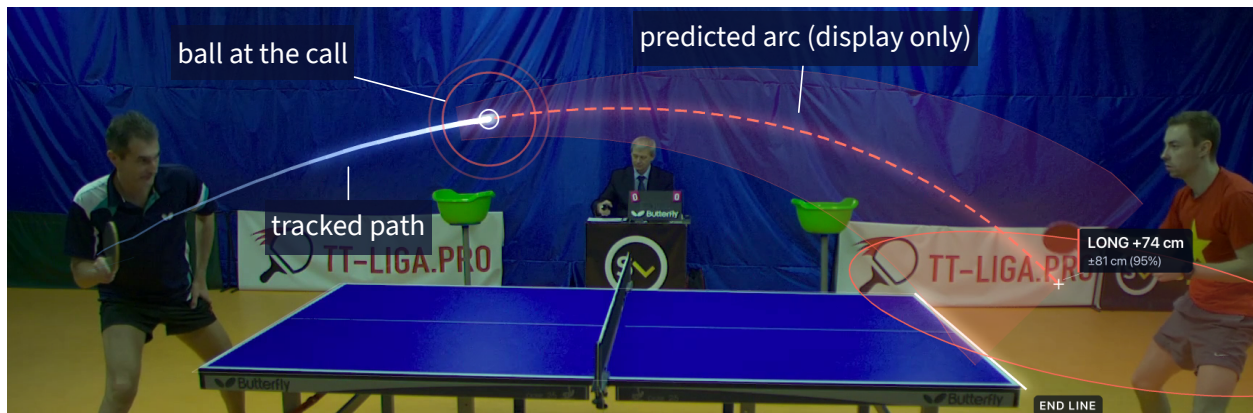


Figure A2. Our tracker on real held-out footage. OpenTTGames test_2 (CC BY-NC-SA 4.0), the frame at the offline rule's call (look-ahead feature); the live causal engine called the same flight later. Tracked path: detections; arc, cone and landing band: physics fit, display only. This is table tennis, not a match feed.

service points: at the first point $\Pr(\text{win}) = 50.0\%$, moving to 51.2% or 47.8%, so $\Delta = 3.4\%$; serving at 5–5, 30–40 in the deciding set, $\Pr(\text{win}) = 33.7\%$, moving to 47.3% or 8.9%: $\Delta = 38.4\%$. Mean $|\Delta|$ over a match: 5.7%.

C11. Latency budget. Eq. 2. With the pre-registered $L = 2.0\text{ s}$, $D = 1\text{ s}$ and our measured $\delta = 0.119\text{ s}$ (pipeline plus network), a point whose book reprices at the median $R = -1.16\text{ s}$ needs $V < -0.28 + \ell$: no feed wins it without an early call. At the 90th percentile $R = 0.08\text{ s}$, $V < 0.96\text{ s}$; the sweep's break-even (1.09 s IS) comes from such late reprices.

C12. Capital and return on capital. Capital = 3× the peak dollars locked in open positions (each locked 4 h, or until resolution). v2 IS: peak \$9,434, capital \$28,302; return = $365 \bar{d} / \text{capital} = 365 \times \$196.24 / \$28,302 = 253\%$ a year.

C13. Capacity and volume share. Capacity is the capital at which the seed-mean Sharpe falls to half its smallest-size value along the size path; share = our notional per day / market volume per day. CV, post hoc, IS, just below the half-Sharpe size: \$10,428 a day against \$4.71M of in-play volume (0.22%) and \$117,047 traded by the fast tier 0–3 s after a point (9%).

Table A2. Latency sweep, every reading: Net \$ per day against feed delay V (s), IS and burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

Stamp lag reading		0	0.5	1	2	3	10	60
pre-registered 2.0 s	IS	89	28	15	-6	-13	-14	-17
	OOS	46	15	4	-11	-13	-13	-17
post hoc 3.14 s	IS	199	133	94	15	-6	-13	-17
	OOS	129	81	57	7	-11	-13	-17
stress 1.0 s	IS	15	-5	-6	-13	-13	-14	-17
	OOS	4	-11	-11	-13	-13	-13	-17
lag 3.0 s	IS	189	121	89	15	-6	-13	-17
	OOS	117	76	46	4	-11	-13	-17
per point, 2.0 s	IS	-30	-17	-17	-17	-17	-17	-17
	OOS	-26	-17	-17	-17	-17	-17	-17
per point, post hoc	IS	199	-45	-17	-17	-17	-17	-17
	OOS	133	-40	-17	-17	-17	-17	-17

Notes: seed means over 20 draws per cell (18,880 runs in all); trade set selected on outcomes, not ex ante. The per-point readings treat the spread of the reprice time as stamp noise, at the pre-registered lag and at the post hoc inference applied point by point. At stamp lag 1.0 s the 1 s cell is -\$6 / -\$11 a day; $V = 0$ reproduces the published tier-0 headline (identical). The model treats the venue hold as a continuous 1.000 s; release on a whole-second clock is not modelled. Source: results/tier0/latency_sweep.json::video_own120.

Table A3. Latency sweep, every reading: Annualised Sharpe against feed delay V (s), IS and burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

Stamp lag reading		0	0.5	1	2	3	10	60
pre-registered 2.0 s	IS	11.4	4.0	2.1	-1.4	-2.6	-2.9	-3.4
	OOS	7.3	2.1	0.3	-2.3	-2.6	-2.6	-3.2
post hoc 3.14 s	IS	22.9	15.5	11.9	2.2	-1.4	-2.7	-3.4
	OOS	19.6	12.2	8.8	0.9	-2.4	-2.6	-3.2
stress 1.0 s	IS	2.1	-1.3	-1.4	-2.6	-2.7	-2.9	-3.4
	OOS	0.3	-2.3	-2.3	-2.6	-2.6	-2.6	-3.2
lag 3.0 s	IS	21.8	14.4	11.4	2.1	-1.4	-2.7	-3.4
	OOS	17.5	11.4	7.3	0.3	-2.3	-2.6	-3.2
per point, 2.0 s	IS	-5.9	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4
	OOS	-5.0	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2
per point, post hoc	IS	25.1	-8.7	-3.4	-3.4	-3.4	-3.4	-3.4
	OOS	21.2	-7.6	-3.2	-3.2	-3.2	-3.2	-3.2

Notes: seed means over 20 draws per cell (18,880 runs in all); trade set selected on outcomes, not ex ante. The per-point readings treat the spread of the reprice time as stamp noise, at the pre-registered lag and at the post hoc inference applied point by point. At stamp lag 1.0 s the 1 s cell is -\$6 / -\$11 a day; $V = 0$ reproduces the published tier-0 headline (identical). The model treats the venue hold as a continuous 1.000 s; release on a whole-second clock is not modelled. Source: results/tier0/latency_sweep.json::video_own120.

Table A4. *Replay on 9 matches' real books (2026-10-03): calling every point or only an ex-ante Markov swing $\geq T$ loses at every V at the pre-registered lag. Net ¢ per share marked at +30 s [match-clustered 95% CI] (fills). Exploratory: the selective variant was added after the all-points replay; one day; assumed feed latency.*

Lag	Points traded	$V = 0$	$V = 0.5 \text{ s}$	$V = 1 \text{ s}$
2.0 s	every point	-0.61 [-0.90, -0.38] (172)	-0.66 [-0.96, -0.41] (167)	-0.94 [-1.28, -0.58] (149)
	swing $\geq 2\text{¢}$	-0.55 [-0.91, -0.22] (140)	-0.61 [-1.02, -0.23] (134)	-0.92 [-1.35, -0.49] (114)
	swing $\geq 4\text{¢}$	-0.37 [-0.75, 0.11] (88)	-0.50 [-0.93, -0.10] (81)	-0.95 [-1.64, -0.40] (63)
	swing $\geq 6\text{¢}$	-0.36 [-0.99, -0.09] (48)	-0.22 [-0.91, 0.20] (47)	-0.26 [-1.35, 0.53] (33)
3.0 s	every point	-0.05 [-0.37, 0.27] (227)	-0.30 [-0.64, 0.03] (208)	-0.66 [-0.96, -0.35] (156)
	swing $\geq 2\text{¢}$	+0.05 [-0.30, 0.47] (197)	-0.25 [-0.65, 0.23] (178)	-0.62 [-1.03, -0.18] (125)
	swing $\geq 4\text{¢}$	+0.55 [0.09, 1.14] (130)	+0.06 [-0.54, 0.79] (119)	-0.42 [-0.92, 0.39] (72)
	swing $\geq 6\text{¢}$	+0.32 [-0.46, 1.51] (69)	-0.21 [-1.31, 1.17] (56)	-0.45 [-0.94, 0.07] (29)

Notes: seed 0; stamp lag 2.0 s pre-registered, 3.0 s an optimistic sensitivity. Source: results/replay/selective/selective.json, research/replay/SELECTIVE.md.

Table A5. *Factor regressions: v2's daily return has no meaningful equity-factor exposure. Fama–French three factors plus momentum, Newey–West 5 lags.*

Specification	α %/day (t)	Mkt (t)	SMB (t)	HML (t)	Mom (t)	R^2	days
IS only (headline)	0.80 (8.5)	0.17 (1.4)	0.08 (0.5)	0.04 (0.4)	0.03 (0.4)	3.3%	138
IS, calendar days	0.68 (9.5)	–	–	–	–	2.3%	206
File (5 OOS days)	0.79 (8.5)	0.17 (1.3)	0.09 (0.5)	0.06 (0.6)	0.03 (0.5)	3.3%	142

Notes: the committed file includes five burned-OOS weekdays and is shown only for completeness. Source: results/alpha/alpha.json::C_factor_neutral; French (2026).

Table A6. *Variant tally: 4,219 configurations tried; the sensitivities chose nothing.*

Family	Count	What it chose
H1-H6 pre-registered and H5/H6 rules	44	frozen primaries (DEVIATIONS D6)
v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi)	3,342	v2 (G_50pct_net100), IS only
v2-safe grid	24	v2-safe (secondary)
tier-0 pre-registered scenarios	432	none (sensitivity grid)
tier-0 v3 in-sample grid	360	frozen v3 rule (blind: FAIL)
maker v1 evaluated variants	8	maker v1 primary (blind: FAIL)
cost-stress cases	4	none (stress)
table-tennis tests TT1-TT5	5	none (untestable)
Total	4,219	
Sensitivity, chose nothing: latency-sweep cells	944	none
Sensitivity, chose nothing: replay cells	36	none
Post hoc presentation choices: headline stamp-lag reading (3.14 s) and the 1 s baseline, fixed after the sweep's burned-OOS cells were computed (peek lines 43, 50, 61–64; replay protocol 0e083ed)	2	post hoc

Source: results/paper/variants.json (each count's source file is listed there).

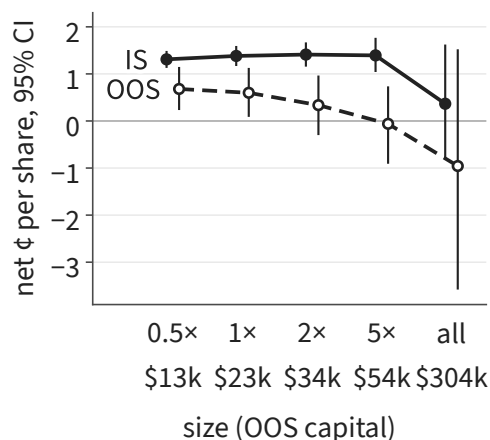
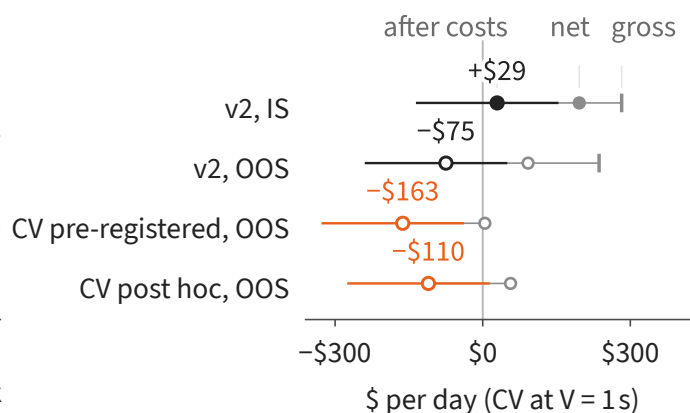
a v2's OOS edge fades as size grows**b** After fixed costs only v2 IS stays positive

Figure A3. Capacity is small and costs bind. (a) v2 net edge by size (caps scaled; optimistic at large size: no impact model); filled = IS, hollow = OOS. (b) \$/day: gross, net of fees, after central fixed costs (bar: low-high); CV rows: the 1 s baseline, burned OOS, pre-registered first; assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

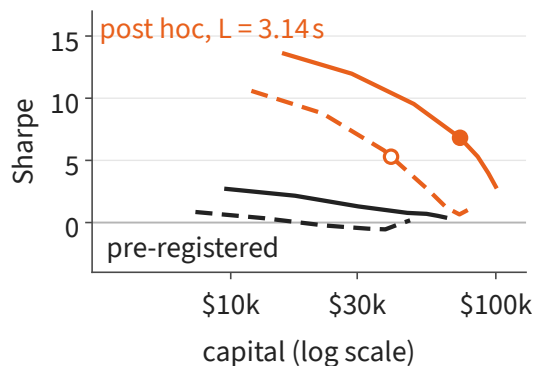
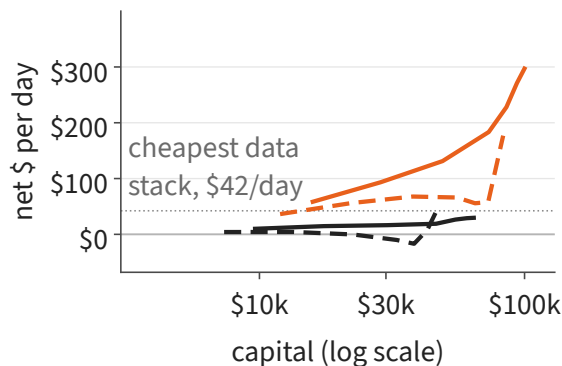
a Sharpe halves by \$40k–\$73k (post hoc)**b** The cheapest data stack needs \$14k–\$16k

Figure A4. Capital capacity of the CV strategy at the 1 s base case. (a) Sharpe and (b) net \$/day against capital along the size path (net cap 50–5,000 shares, \$250 orders, 10 matches a day, half the stale depth); circles: where the post hoc Sharpe halves; dotted: the cheapest data stack. Solid = IS, dashed = burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured. Source: results/capacity.

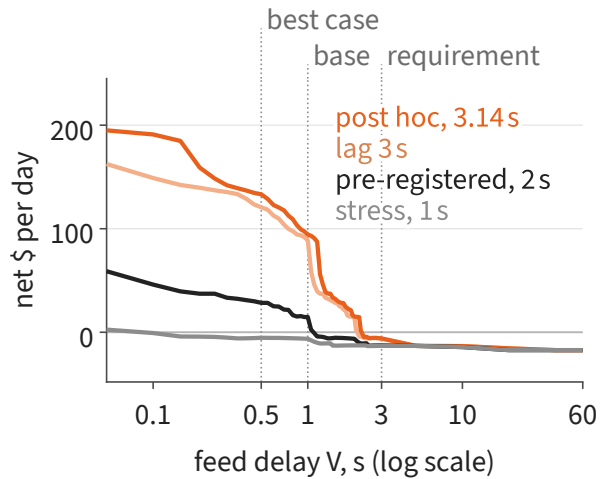
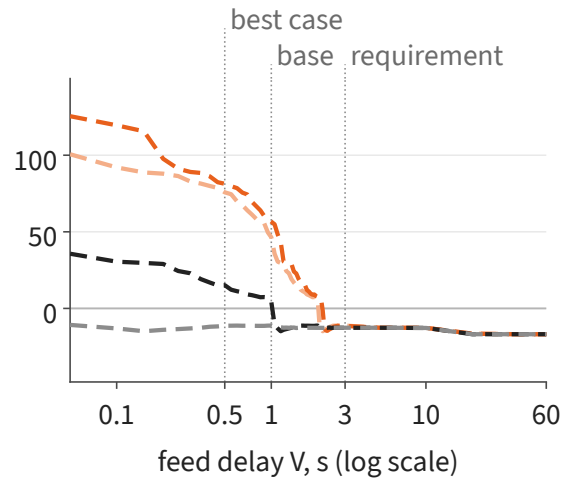
a IS: break-even moves with the lag**b** OOS: the same shift

Figure A5. The break-even shifts one for one with the stamp lag. Net \$/day against feed delay for every stamp-lag reading, (a) IS and (b) burned OOS; dotted: the three scenarios. Simulated: assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0/latency_sweep.csv.

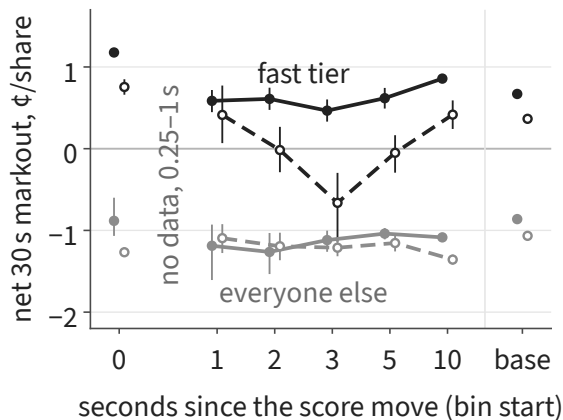
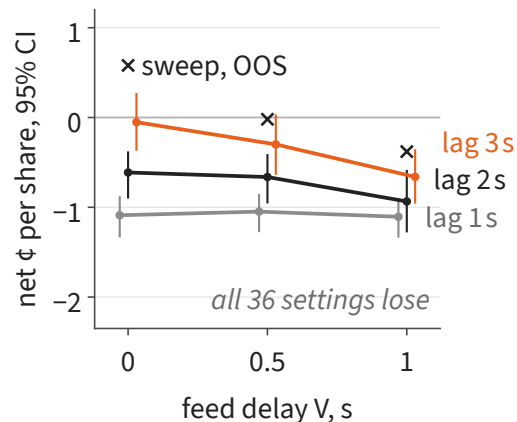
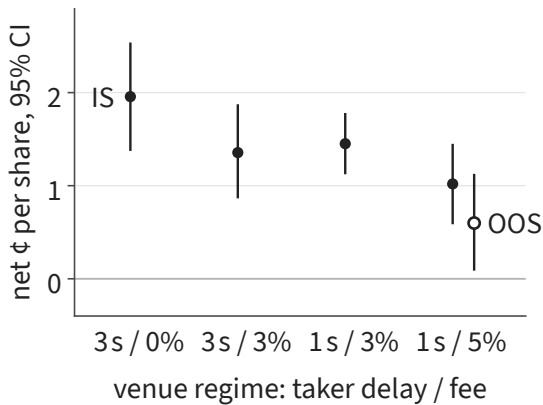
a The fast tier stays ahead at every horizon**b** Trading every point loses at every delay

Figure A6. The fast tier leads at every horizon; trading every point on real books loses. (a) Net 30s markout by seconds since the score move; filled = IS, hollow = OOS. (b) Replay of 9 matches recorded live on 2026-10-03 on their real order books, every point called (model leads, Florida network); net \$ per share marked at +30s, match-clustered 95% CI; x: the sweep's OOS cell. One day; assumed feed latency (licensed feed not purchased); paper only. Source: results/{decay,replay}.

a Edge stays positive in every venue regime



b 7 of 7 in-sample months are positive

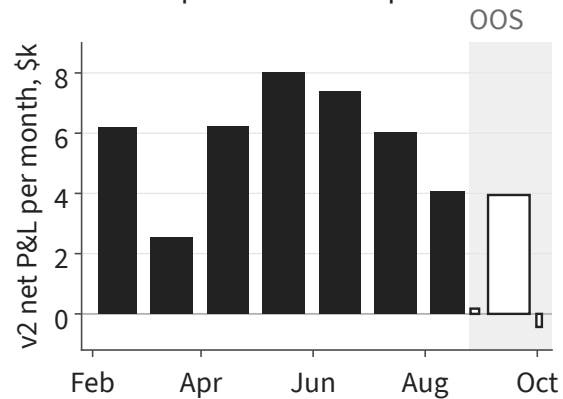


Figure A7. v2's edge falls as the venue's protection falls, and stays positive in every regime. (a) Net ¢ per share by venue regime (taker delay / fee), 95% CIs; (b) net P&L by month. Measured at the fast tier's own fills. Source: results/risk/risk_stats.json.

Only v2 at fast-tier fills clears zero

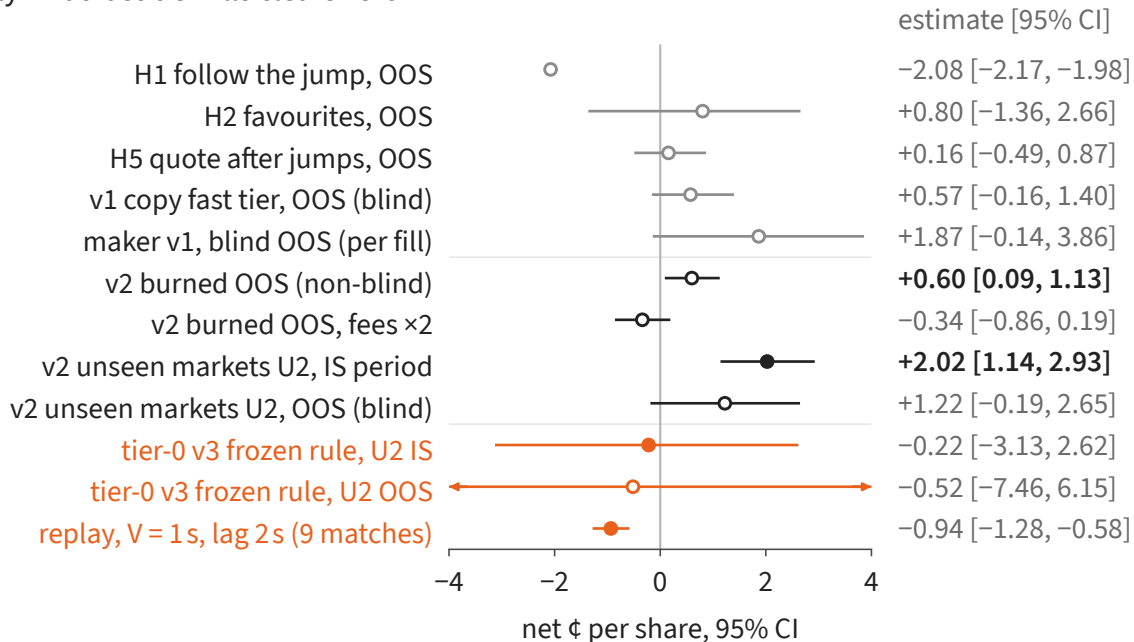


Figure A8. Every test on one axis. Point estimates and 95% CIs, net ¢ per share; hollow = an OOS test; a CI above zero is set in semibold; arrowheads: the CI runs past the axis. Source: as Table 3.

Five wallets carry 81% of in-sample P&L

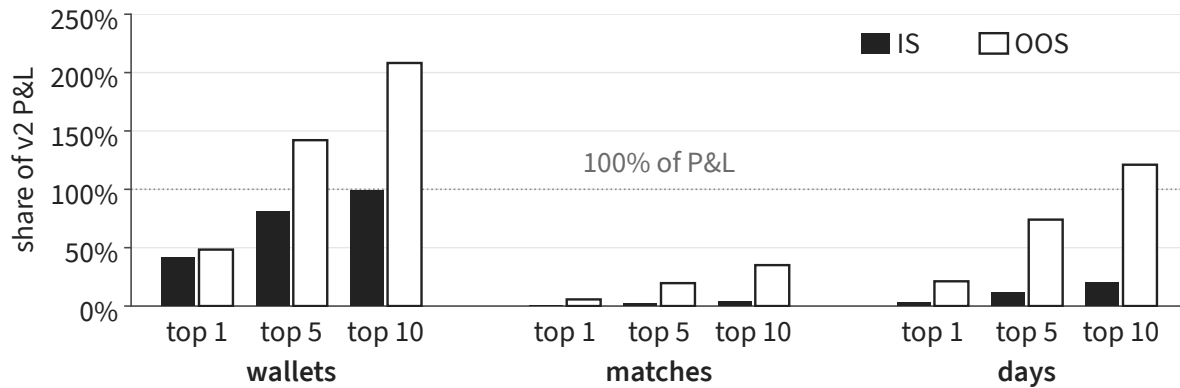
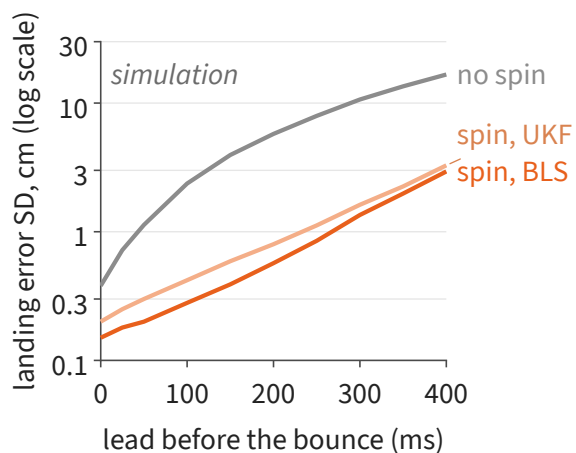


Figure A9. *v2 P&L is concentrated in a few copied wallets.* Share of v2 P&L carried by the top 1, 5 and 10 wallets, matches and days (above 100% means the rest lost money). Source: results/alpha/alpha.json.

a Spin-aware fit cuts error 10× at 200 ms



b OUT calls: precision 1, recall ≥ 0.88

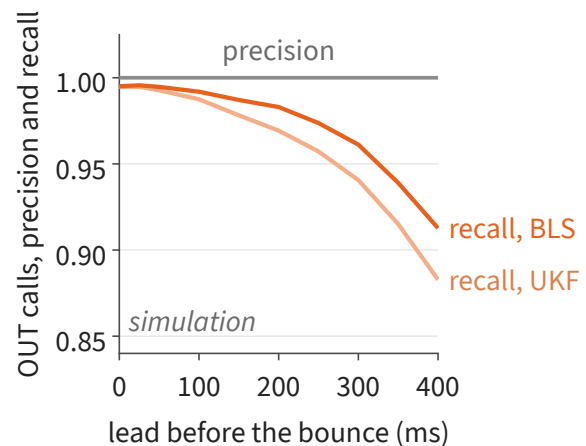


Figure A10. *Simulation: a spin-aware tennis filter cuts landing error.* Landing-point error and OUT-call accuracy against lead before the bounce for a 340 fps, 3.6 mm-noise ball tracker (simulation; no tennis video was used). Source: results/spin/tennis/key_numbers.json.

Table A7. Fixed-cost assumptions behind Fig. A3b (\$ per month unless stated).

Item	Low	Central	High	Label
Official live ATP/WTa point-data feed licence	\$1,250	\$5,000	\$10,000	ASSUMPTION
London gateway VPS (AWS eu-west-2, on-demand Linux)	\$34	\$77	\$98	ESTIMATE
Polymarket CLOB / data API	\$0	\$0	\$0	ESTIMATE
HiPerGator (training, backtests)	\$0	\$0	\$0	ESTIMATE
Courtside camera operator (tier-0 only)	\$36	\$36	\$36	ESTIMATE
Cloud GPU for CV inference, per covered event hour (tier-0 only)	\$1	\$1	\$1	ESTIMATE
Operator + GPU hours per covered match (tier-0 only)	\$3	\$4	\$5	ASSUMPTION
Camera position / venue access fee per event (tier-0 only)	\$0	\$250	\$500	ASSUMPTION
120 fps camera kit, amortised over 24 months (tier-0 only)	\$648	\$1,000	\$2,000	ESTIMATE

Notes: no public price exists for an official low-latency tennis feed; the range is anchored to reported figures (bases and URLs in the source file). Source: results/financials/financials.json::cost_assumptions.

Table A8. *The latency ladder: method and source of every row of Fig. A1a.*

Source	Delay	Method and citation
Our pipeline, frame to order-ready	54 ms	measured p50, own footage over WebRTC, laptop, order not sent (results/e2e)
Our WebRTC → CV call	46 ms p50, 67 ms p99	measured: own clip over loopback (55 calls; engine fed below real time so it keeps up); video leg 4–8 ms
Our CV engine, call-ready (p50–p99)	4.6–12.2 ms	measured, one L4 GPU, 120 fps held-out video
Network, Florida → venue gateway	65 ms	measured RTT/2 at the e2e orders (live feed p50 65 ms)
Venue taker order delay	1 s	venue rule (Polymarket, 2026a)
Official umpire stamp after the point	1–3 s	unmeasured model grid (Regen Sports, 2026; Stats Perform, nd)
Book reprice vs official stamp	–1.2 s	measured median, $n = 482$
Licensed betting video	0.5–8 s	vendor claims, unverified, none for tennis (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Ably, nd)
Fastest public stream of a listed match	12.2 s	measured, preliminary
ESPN / PM sports / WTA vs book	28.2 / 30.0 / 44.1 s	measured medians, 2026-10-03 (ESPN, 2026; WTA, 2026)
TV and public streams, general	0.9–60 s	published comparisons (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026)
Source: research/v2/latency/results.json, results/tier0/latency_sweep.json::sources, re- sults/home_stream/sub_second_routes.json, results/engine/online_vs_offline.json, re- sults/{e2e,webrtc}.		

Table A9. Every logged read of held-out data (77 lines of results/oos_peek.log at build time), with its class; a line logged after its run is set in bold.

UTC	Class	What was read (truncated)
10-03 10:42	other	OOS prints built
10-03 13:24	non-blind	v2 evaluated on burned OOS (non-blind, labelled)
10-03 13:53	non-blind	v2-causal (D9 fix) evaluated on burned OOS (non-blind)
10-03 17:23	non-blind	tier0 counterfactual: OOS jump table built (burned OOS, non-blind)
10-03 17:29	non-blind	tier0 counterfactual grid evaluated on burned OOS (non-blind, labelled)
10-03 17:31	blind first run	U2 out-of-universe test evaluated (PREREG research/v2/expand, first run)
10-03 17:37	non-blind	v2-safe (lowloss PREREG) evaluated on burned OOS (non-blind, labelled)
10-03 17:37	blind first run	v2-safe U2 out-of-universe test evaluated (PREREG research/v2/lowloss, first run)
10-03 17:43	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 17:45	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 17:48	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 18:00	non-blind	v2 cost stress (fees/costs doubled, book fixed) on burned OOS (non-blind, no parameter change)
10-03 18:06	non-blind	tier0 corrections: OOS post-jump prices built (burned OOS, non-blind, labelled)
10-03 18:13	non-blind	tier0 counterfactual (verifier corrections) evaluated on burned OOS (non-blind, labelled)
10-03 18:27	descriptive	maker venue rules: descriptive in-play book spread/depth + tick changes read from data/live_v2 forward recordings (no strategy, no P&L, no parameter c...
10-03 18:42	non-blind	financials: v2 / v2-safe size-scaled variants (0.5x, 2x, 5x, all prints) evaluated on burned OOS (non-blind, capacity curve only)
10-03 18:47	descriptive	maker live engine plumbing: data/live_v2 clob_17/clob_13 read for message ordering only (price_change vs last_trade_price timestamps, trade dedupe); n...
10-03 18:49	blind first run	maker v1 blind OOS evaluated (first run) (PREREG research/v2/maker/PREREG.md (A); first use of OOS side-market tapes; burned OOS moneyline tapes used ...
10-03 18:49	audit	maker v1 OOS reproduction (recompute committed results.json; no parameter change)
10-03 19:00	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_11.jsonl.gz,data/live_v2/clob_20261003_1123_2026100...
10-03 19:13	live	maker v1 live paper TEST run (not the pre-registered session) START: run=20261003T190317Z commit=56a9c8f77b1aaf7b1306c2f1b06d6a2e05c798a8+dirty until=...

UTC	Class	What was read (truncated)
10-03 19:15	audit	P17 clean-clone reproduce.sh (HEAD 56a9c8f, scratch clone, data/linked, backtest workers=1): re-runs the committed frozen scripts incl. run_all.py --...
10-03 19:19	live	maker v1 live paper engine REPLAY (B2: own raw log) on data/live_maker/raw_20261003T190317Z.jsonl.gz window=-+Nonemin clock=recv: run=20261003T191913Z...
10-03 19:20	non-blind	PM review P07: wallet-clustered bootstrap CI of the frozen v2 causal book on burned OOS (non-blind, labelled; the CI the note already quotes, no rule ...
10-03 19:23	blind first run	table tennis TT1-TT4 evaluated (first run)
10-03 19:25	descriptive	TT5 signal decay (scripts/signal_decay.py, commit a5769c786f935dcc05ff2ab90a1947ad0cde4cb5+dirty(signal_decay.py)): reads table tennis UTT OOS prints ...
10-03 19:28	other	table tennis post-hoc diagnostics after TT1-TT4 (closing-print size, fast-tier distance; not pre-registered, no verdict change)
10-03 19:28	live	maker v1 live paper TEST run (not the pre-registered session) START: run=20261003T191915Z commit=ad164da6da24325afcfdbab8fe24dcf5f0516fed+dirty until=...
10-03 19:58	live	maker v1 live paper session (B) START: run=20261003T193534Z commit=114c722e6b203421d8b4f55a5da2b54a1658b658 until=2026-10-04T11:30:00.000Z (PREREG res...
10-03 20:02	audit	audit: independent TT3 reproduction (research/tt/audit_tt3.py; reads UTT IS+OOS tapes/prints, already evaluated in 2195e86) plus labelled post-hoc dia...
10-03 20:03	audit	audit: table tennis checks (research/tt/audit_checks.py): provenance, fees/delays, voids, counts, liquidity; reproduces TT1 closing bins (IS+OOS, alre...
10-03 20:04	audit	audit: financials/risk audit (scripts/financials_audit.py) re-reads the frozen v2 burned-OOS trades and maker blind-OOS fills to reproduce numbers alr...
10-03 20:07	audit	audit: RE-RUN of research/tt/audit_tt3.py with three added post-hoc counts (closest wallet's share of TT2 others' prints, 0-3 s prints at fav >= 0.95 ...
10-03 20:07	audit	audit: RE-RUN of research/tt/audit_tt3.py (traceback location recorded as repo-relative src/v2.py frames only; same reads and numbers, no rule or verd...
10-03 20:09	descriptive	alpha breakdown: descriptive splits of existing v2 trades incl. burned OOS (no parameter change)
10-03 20:10	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[12].jsonl.gz window=2026-10-03T12:00:00.000Z+60.0...
10-03 20:10	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[12].jsonl.gz window=2026-10-03T12:00:00.000Z+60.0...
10-03 20:11	audit	audit: maker v1 blind OOS (A) independent recompute (own code in scratchpad, no import of scripts/maker_oos.py or research/v2/maker/oos_*.py); reads O...

UTC	Class	What was read (truncated)
10-03 20:13	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:13	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[1234].jsonl.gz window=2026-10-03T14:00:00.000Z+12...
10-03 20:14	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_11.jsonl.gz window=2026-10-03T11:35:00.000Z+25.0min...
10-03 20:14	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[56789].jsonl.gz window=2026-10-03T16:00:00.000Z+2...
10-03 20:16	non-blind	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen)
10-03 20:16	audit	audit: TT5 same-block split and v2 trade join (research/decay/audit_sameblock.py): re-reads tennis U1 IS + burned OOS prints (non-blind, already evalu...
10-03 20:18	audit	audit: selection check for maker v1 OOS (A): data-api /trades counts and notional (no prices used, no signal, no P&L) for a random sample of 60 OOS si...
10-03 20:21	audit	audit: maker v1 OOS (A) stress (viii) alternative fee reading (makerBaseFee 1000 bps x min(px,1-px), CTF-exchange form) on the already-evaluated book...
10-03 20:22	audit	audit: provenance spot-check of maker v1 OOS cache: re-download 8 cached OOS side tapes (data-api) and 1 Gamma events batch and compare with the cache...
10-03 20:22	live	maker v1 live paper engine REPLAY (B2: own raw log) on data/live_maker/raw_20261003T191915Z.jsonl.gz window=-+Nonemin clock=recv: run=20261003T202228Z...
10-03 20:22	live	maker v1 live paper engine REPLAY (B2: own raw log) on /private/tmp/claude-501/-Users-ojasvamishra32-Library-Application-Support-Claude-scratch-worksp...
10-03 20:24	non-blind	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen)
10-03 20:25	audit	tier0 v3 blind test (a) burned OOS START (scripts/tier0_v3_blind.py; PREREG research/v2/tier0_v3/PREREG.md section 5; non-blind window seen by v1/v2/t...
10-03 20:26	blind first run	tier0 v3 blind test (b) U2 START (scripts/tier0_v3_blind.py; PREREG research/v2/tier0_v3/PREREG.md section 4; first tier-0 table on U2: jump tables + ...
10-03 20:28	blind first run	tier0 v3 blind tests (a)+(b) DONE: burned OOS FAIL, U2-IS FAIL, U2-OOS FAIL -> results/tier0_v3/blind.json
10-03 20:29	blind first run	tier0 v3 blind tests (a)+(b) DONE: burned OOS FAIL, U2-IS FAIL, U2-OOS FAIL -> results/tier0_v3/blind.json
10-03 20:37	audit	audit: tier-0 v3 blind tests adversarial audit (independent code in scratchpad; COUNTERFACTUAL): re-reads saved v3 burned-OOS/U2 outputs (results/tier...

UTC	Class	What was read (truncated)
10-03 20:40	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:45	audit	independent rebuild: tier-0 frozen v3 re-implemented in scratchpad (no scripts/tier0_v3_* import; COUNTERFACTUAL) reads U2 prints/universe/raw tapes, ...
10-03 20:46	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:52	live	maker v1 live paper session (B) DEVIATION L11-L14 (research/v2/maker/DEVIATIONS_LIVE.md): session run=20261003T193534Z PID 65274 stopped by SIGTERM at...
10-03 21:05	descriptive	alpha breakdown: descriptive splits of existing v2 trades incl. burned OOS (no parameter change)
10-03 20:17	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44606812; line written to the cluster copy of this log, s...
10-03 20:36	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44608194, the run behind commit f5ff9ff; line written to ...
10-03 20:53	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [audit: independent recompute of sweep cells incl. burned OOS video V = 1...(LOGGED AFTER THE RUN)
10-03 20:59	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44610737, audit re-run adding stamp-lag cells; line writt...
10-03 21:14	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [logged before the replay...
10-03 21:56	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [visuals: scripts/match_r...(LOGGED AFTER THE RUN)
10-03 22:18	non-blind	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py: local v2 1x check, then the HiPerGator ...
10-03 22:24	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [audit: scripts/match_rep...
10-03 23:01	live	match replay selective variant: exploratory, post hoc (forward live recordings of 2026-10-03) [scripts/match_replay_selective.py: the replay restricte...
10-03 23:17	audit	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [audit of commit f9d8f94: independent recompute of 5 grid cells inc...
10-03 23:18	non-blind	redteam causal-CV cell (INTEGRATION_TODO C5; scripts/redteam_causal_cv.py): own120_engine = live causal engine recall/precision at V 0-3 s under tourn...

UTC	Class	What was read (truncated)
10-03 23:23	non-blind	backtest example match (video asset): latency-sweep rerun at V=1 s, calibrated reading, read burned OOS prints for per-match P&L (non-blind, illustrat...
10-03 23:24	live	match replay selective variant: exploratory, post hoc (forward live recordings of 2026-10-03) [audit of commit b20a4e4: adversarial check of scripts/m...
10-04 00:28	non-blind	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [fix pass after the capacity and e2e verifiers: pre-flight test of ...
10-04 00:28	audit	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py --grid on Ojasvas-MacBook-Air-3.local, r...
10-04 00:36	audit	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py --grid --resume on Ojasvas-MacBook-Air-3...
10-04 02:02	live	maker v1 live paper session (B) START: run=20261003T204812Z commit=c655981f378a6a5567eef4470c2bef19b1e4f9e7 until=2026-10-04T11:30:00.000Z (PREREG res...

Table A10. *Pre-registration and deviation trail (git history of the hypothesis, protocol and deviation files).*

Commit	UTC	Subject
a5769c7	19:24 (Oct 3)	tier-0 PREREG: its text says written 17:25 UTC; committed with its results; OOS grid first run 17:29 UTC (peek log)
7232986	2026-10-03 09:50	Pre-register hypotheses before any results
0f1cc89	2026-10-03 10:15	Data pipeline, Markov model, physics sim, strategies, tier and fast-tier analysis
9d61d1b	2026-10-03 10:32	Freeze all strategy settings before out-of-sample evaluation
69a675f	2026-10-03 10:48	Out-of-sample evaluation, diagnostics, figure fixes
be64ea4	2026-10-03 12:00	Ball tracking results (table tennis H3, tennis broadcast), interactive visual, live-book paper engine
1f229bc	2026-10-03 13:22	Pre-register v2 rule and forward test; add v2 research lenses and verifications
845666f	2026-10-03 13:55	Fix onset hindsight: causal v2; amend forward-test pre-registration before any forward data
e7f34d8	2026-10-03 13:56	Forward window starts 14:00 UTC, after the amended pre-registration
8ede061	2026-10-03 15:00	Pitch deck (12 slides, reviewed); note framing fixes; D10 corrections
337bf7c	2026-10-03 17:12	Pre-register out-of-universe test of frozen v2 (U2)
342d515	2026-10-03 17:30	Pre-register v2-safe (loss-averse, chosen on IS only)
c29734b	2026-10-03 18:26	Pre-register maker v1 blind OOS test and live paper session
0f01362	2026-10-03 18:31	Pre-register table tennis study (TT1-TT5) before fetching tapes

Commit	UTC	Subject
a5769c7	2026-10-03 19:24	Tier-0 CV-edge counterfactual (assumed licensed feed + courtside camera; measured parameters): pre-registration, original run, two verifier reports, revision with live-book fill pricing; original kept as record
ba0c3d4	2026-10-03 20:16	Pre-register frozen tier-0 v3 and its blind tests
0e083ed	2026-10-03 21:14	Match replay protocol: selection rule and per-point procedure, written before any P&L
12260cd	2026-10-04 02:18	Stop the live paper session and drop the forward test by team decision (disclosed: HYPOTHESIS_V2 A3, DEVIATIONS_LIVE L15)
8156177	2026-10-04 02:59	Reinstate the pre-registered forward test before any forward data is read (HYPOTHESIS_V2 A4)